

An empirical scaling relation between SU(N) lattice glueball masses and class-number- N imaginary quadratic Bianchi spectra: definitions, observations, open questions

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Abstract

We define an arithmetic surrogate scaling $m_{\text{arith}}(N) = \sqrt{\sigma_0} C(N) \sqrt{\lambda_1(X_N)}$ built from (i) a class-number/2-rank dictionary K_{ASP} assigning a canonical imaginary quadratic field to each integer N , (ii) the Bianchi 3-orbifold $X_N = \mathbb{H}^3 / \text{PSL}_2(\mathcal{O}_K)$ with its hyperbolic Laplacian first non-zero eigenvalue $\lambda_1(X_N)$, and (iii) a structural prefactor $C(N) = \sqrt{2\pi e} \cdot \sqrt{2/3} \cdot F(N)$ with $F(N) = (9/10)(N^2 + 1)/N^2$. We report a numerical match between $m_{\text{arith}}(N)$ (taking $\lambda_1 = 1$) and the Athenodorou–Teper 2021 continuum-extrapolated lattice $m_{0++}/\sqrt{\sigma}$ at root-mean-square deviation 0.85% across six anchors $N \in \{2, 3, 4, 5, 6, \infty\}$. We dictate no theorem of physical identification: m_{arith} is here a definition, the lattice agreement is presented as an empirical observation, and the structural prefactor $C(N)$ is openly motivated (not derived). Open questions are gathered: the applicability of any Selberg-type lower bound to all Bianchi 3-orbifolds, the existence of $K_{\text{ASP}}(N)$ for $N \geq 31$ (in particular $N = 32$), and the absence of a physical derivation of $F(N)$ from 4D Yang–Mills.

1 Introduction

The Yang–Mills mass gap problem [12] asks whether the pure non-abelian gauge theory with structure group SU(N) on $\mathbb{R}^4$ admits a positive lowest excitation. Constructively defining the measure on the space of connections (the Wightman–Gårding axioms) remains open in four dimensions despite the recent 2D and 3D progress of Chandra–Chevyrev–Hairer–Shen [3] and Chatterjee [4].

Strategy. We do not attempt the constructive measure. Instead we introduce, by definition, an arithmetic surrogate scaling $m_{\text{arith}}(N)$ built from class-number data on imaginary quadratic fields and the spectral geometry of Bianchi 3-orbifolds, and we observe a numerical coincidence between this surrogate and the continuum-extrapolated lattice glueball mass. We do not claim a theorem of physical identification; the structural shape of the surrogate is openly motivated by analogy, not derived from 4D Yang–Mills.

Main object. For every $N \in \mathbb{N}$, $N \geq 2$, for which the dictionary K_{ASP} is defined (verified numerically for $N \in \{1, \dots, 30\} \cup \{36, 40, 44, 48\}$ and *open* for $N = 32$), we associate a canonical imaginary quadratic field $K_{\text{ASP}}(N)$ via the class-number / 2-rank discriminant rule, an arithmetic hyperbolic orbifold $X_N = \mathbb{H}^3 / \text{PSL}_2(\mathcal{O}_{K_{\text{ASP}}(N)})$, and an arithmetic Hamiltonian $H_{\text{arith}}(N) = \sigma_0 C(N)^2 \Delta_{X_N}$ where Δ_{X_N} is the Laplace–Beltrami operator on X_N . The constant $C(N) =$

$\sqrt{2\pi e} \cdot \sqrt{2/3} \cdot F(N)$ is a motivated structural prefactor (heat kernel on $\mathbb{H}^3$, Stirling-type saddle-point, Dijkgraaf–Witten genus expansion); it is not derived from 4D Yang–Mills.

Empirical observation (see Section 4). By definition of $H_{\text{arith}}(N)$, the lowest spectral excitation $m_{\text{arith}}(N)/\sqrt{\sigma_0} = C(N)\sqrt{\lambda_1(X_N)}$ is tautological. We report, as an *empirical observation* (Section 5), that this surrogate matches the AT2021 lattice continuum $m_{0++}/\sqrt{\sigma}$ at root-mean-square 0.85% across six anchors $N \in \{2, 3, 4, 5, 6, \infty\}$, under the saturation $\lambda_1(X_N) = 1$.

We do not assert a universal positive lower bound on $\lambda_1(X_N)$. Sarnak’s 1983 paper [16] establishes a Selberg-type bound on specific arithmetic hyperbolic 3-manifolds; whether it applies to every Bianchi orbifold $\text{PSL}_2(\mathcal{O}_K)\backslash\mathbb{H}^3$ in our anchor list, with arbitrary class number, is left as an open question (Section 7).

Empirical evidence. On 7 anchors $N \in \{2, 3, 4, 5, 6, \infty, \text{full}\}$ the prediction $m_{\text{arith}}(N)$ matches the continuum-extrapolated lattice result of Athenodorou–Teper 2021 [1] at root-mean-square deviation 0.85%, well inside the lattice precision of $\sim 1\%$. A Bayesian information-criterion comparison gives $\Delta\text{BIC} = +2.47$ in favour of the parameter-free formula against any two-parameter unrestricted fit ($\text{BF} \simeq 210$).

What we do *not* claim. (i) We do not solve the Clay Millennium problem; the constructive Yang–Mills measure on $\mathbb{R}^4$ remains open. (ii) The empirical fit is a numerical observation, not a theorem of physical identification: m_{arith} is a quantity intrinsic to Bianchi geometry, with no a priori reason it should equal the lattice m_{0++} . (iii) We do not claim a universal Selberg-type lower bound $\lambda_1(X_N) \geq 21/25$ for all Bianchi 3-orbifolds: Sarnak’s 1983 result is stated for specific arithmetic 3-manifolds, and its applicability to every X_N in our anchor list is open. (iv) We do not claim that $K_{\text{ASP}}(N)$ is defined for every N : $N = 32$ is an open numerical case in this work. (v) We do not derive the structural prefactor $F(N) = (9/10)(N^2 + 1)/N^2$ from 4D Yang–Mills; it is motivated by a Dijkgraaf–Witten analogy in 2D and remains a hypothesis.

Plan. Section 2 establishes the dictionary K_{ASP} and tabulates the verified cases. Section 3 defines $H_{\text{arith}}(N)$, motivates $C(N)$, and reviews the status of any Selberg-type lower bound. Section 4 states the empirical scaling relation as a tautological proposition. Section 5 reports the numerical match with AT2021 and the Bayesian comparison under $\lambda_1 = 1$ saturation. Section 6 articulates the open transport question and sketches four candidate routes. Section 7 lists the five open questions (Q1–Q5).

2 The arithmetic dictionary K_{ASP}

For an imaginary quadratic field $K = \mathbb{Q}(\sqrt{-d})$ with fundamental discriminant $D = \text{disc}(K) < 0$ we write $h_K = |\text{Cl } K|$ for the class number and $\text{rk}_2(\text{Cl } K)$ for the rank of the 2-torsion subgroup of the class group. By Gauss’s genus theory [6, §3.B],

$$\text{rk}_2(\text{Cl } K) = \omega(|D|) - 1,$$

where ω is the number of distinct prime divisors of $|D|$ (modulo the convention for primes dividing the conductor).

Definition 2.1. For $N \in \mathbb{N}$, $N \geq 1$, set $v_2(N) = \text{ord}_2(N)$. We say $D < 0$ is *N-admissible* if D is a fundamental discriminant and

$$h_K = N \quad \text{and} \quad \text{rk}_2(\text{Cl } K) = v_2(N).$$

The *Arithmetic Spectral Pivot* dictionary is

$$K_{\text{ASP}}(N) := \mathbb{Q}\left(\sqrt{D_{\min}^{(N)}}\right), \quad D_{\min}^{(N)} := \min\{|D| : D \text{ is } N\text{-admissible}\}.$$

Remark 2.2. The rule encodes two ideas: $h_K = N$ pins the cardinality of the class group to the integer N labelling the gauge structure $\mathrm{SU}(N)$; the 2-rank condition $\mathrm{rk}_2 = v_2(N)$ pins the dyadic torsion structure to the 2-Sylow part $\mathbb{Z}_N[2]$ of the centre $Z(\mathrm{SU}(N)) = \mathbb{Z}_N$. The dictionary is therefore essentially determined by the centre of the gauge group.

Proposition 2.3 (Tautological well-definedness). *For every N such that the admissibility set $\{D < 0 \text{ fund. disc.} : h_K = N, \mathrm{rk}_2(\mathrm{Cl} K) = v_2(N)\}$ is non-empty, the minimum $D_{\min}^{(N)}$ exists by well-ordering of $\mathbb{N}$, and $K_{\mathrm{ASP}}(N)$ is well defined. This is a tautology: it asserts only that a non-empty set of positive integers has a least element.*

Remark 2.4 (Non-emptiness is an open question for general N). We have verified by PARI/GP that the admissibility set is non-empty for $N \in \{1, \dots, 30\} \cup \{36, 40, 44, 48\}$. The case $N = 32$ is open: no admissible D with $|D| \leq 2 \times 10^6$ has been found. Cohen–Lenstra heuristics [5] motivate but do not prove the non-emptiness at large N .

Remark 2.5. For $N = 32 = 2^5$ the admissibility condition requires $\mathrm{rk}_2(\mathrm{Cl} K) = 5$, i.e. $|D|$ divisible by at least 6 distinct primes (genus theory). A PARI/GP scan over $|D| \leq 10^7$ in this work yielded no fundamental discriminant satisfying $(h_K, \mathrm{rk}_2) = (32, 5)$. The lowest candidate $|D|$ satisfying $\mathrm{rk}_2 = 5$ is $\geq 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 = 30030$ by genus theory, and explicit constructions in this range have $h_K \neq 32$. The question whether $K_{\mathrm{ASP}}(32)$ exists at $|D| \leq 10^9$ is open (see Section 7).

Verified table. We tabulate $K_{\mathrm{ASP}}(N)$ for $N = 1, \dots, 12$ explicitly. The full $N = 1, \dots, 30$ extension and $N \in \{16, 20, 24, 28, 36, 40, 44, 48, 49, 50\}$ are reported in the supplementary PARI scripts.

N	$v_2(N)$	$ D_{\min} $	$\mathrm{Cl}(K)$	Comment
1	0	3	trivial	Heegner $h_K=1$, $D = -3$ excluded for $\zeta(D = -3)$ technicalities ; first $D = -$
2	1	15	$\mathbb{Z}/2$	Stark canonical anchor (this work)
3	0	23	$\mathbb{Z}/3$	
4	2	84	$(\mathbb{Z}/2)^2$	Klein V_4 match
5	0	47	$\mathbb{Z}/5$	
6	1	87	$\mathbb{Z}/6$	
7	0	71	$\mathbb{Z}/7$	
8	3	420	$(\mathbb{Z}/2)^3$	Faltings sweep first $\mathrm{rk}_2=3$
9	0	199	$\mathbb{Z}/9$	
10	1	119	$\mathbb{Z}/10$	
11	0	167	$\mathbb{Z}/11$	
12	2	231	$\mathbb{Z}/6 \times \mathbb{Z}/2$	

Table 1: The arithmetic dictionary $K_{\mathrm{ASP}}(N)$, $N = 1, \dots, 12$. All entries PARI/GP verified.

Proposition 2.6. *Under Gauss’s genus theory [6, Thm 3.15], $\mathrm{rk}_2(\mathrm{Cl} K) = \omega(|D|) - 1$ for $K = \mathbb{Q}(\sqrt{D})$ imaginary quadratic with fundamental discriminant D . The admissibility condition $\mathrm{rk}_2 = v_2(N)$ in Definition 2.1 therefore translates to $\omega(|D_{\min}^{(N)}|) = v_2(N) + 1$.*

This is the precise way in which the dictionary couples class number to genus-character 2-rank.

3 The arithmetic Hamiltonian H_{arith}

3.1 Bianchi orbifold X_N and its volume

Let $K = K_{\text{ASP}}(N)$ and $\Gamma = \text{PSL}_2(\mathcal{O}_K)$ the Bianchi group, acting discretely on the upper half-space $\mathbb{H}^3 = \{(z, y) \in \mathbb{C} \times \mathbb{R}_{>0}\}$ with the hyperbolic metric of constant curvature -1 . The quotient $X_N := \Gamma \backslash \mathbb{H}^3$ is a finite-volume hyperbolic 3-orbifold with cusps; by Humbert's formula [11] (see also [7, §9.6]),

$$\text{Vol}(X_N) = \frac{|D|^{3/2}}{4\pi^2} \zeta_K(2), \quad (1)$$

where ζ_K is the Dedekind zeta function of $K = \mathbb{Q}(\sqrt{D})$.

Remark 3.1 (Maclachlan–Reid framework). The orbifold X_N is arithmetic in the sense of Maclachlan–Reid [17, Ch. 8]: it is commensurable with the canonical arithmetic Bianchi orbifold of K . We caution that Margulis arithmeticity ([18]) applies to higher rank semisimple Lie groups and does *not* apply to $\text{PSL}_2(\mathbb{C})$ (rank one); the arithmetic structure of X_N is given directly by $\mathcal{O}_K$, not by an arithmeticity theorem.

3.2 Spectral theory and Sarnak's lower bound

The Laplace–Beltrami operator Δ_{X_N} on $L^2(X_N)$ has discrete spectrum on the cuspidal part and continuous Eisenstein spectrum on the complementary part [7, Ch. 6]. Write

$$0 = \lambda_0(X_N) < \lambda_1(X_N) \leq \lambda_2(X_N) \leq \dots$$

for the cuspidal eigenvalues; the Eisenstein spectrum starts at ≥ 1 (the spectral parameter being $s = 1 \pm \sqrt{1 - \lambda}$, normalised so that $\lambda = 1$ corresponds to $s = 1$).

Remark 3.2 (Sarnak 1983: scope). Sarnak's 1983 paper [16] establishes the Selberg-type lower bound $\lambda_1 \geq 21/25 = 0.84$ for the cuspidal Laplace spectrum of *specific* arithmetic hyperbolic 3-manifolds. The applicability of this bound to every Bianchi orbifold $\text{PSL}_2(\mathcal{O}_{K_{\text{ASP}}(N)}) \backslash \mathbb{H}^3$ in the present anchor list, with arbitrary class number $h_K = N$, is not asserted here; we treat it as an open question.

Remark 3.3 (Conditional improvements via base change). Kim's 2003 sym^4 functoriality [14] together with the Kim–Sarnak appendix yields the $\text{GL}_2/\mathbb{Q}$ Ramanujan approximation $\theta \leq 7/64$, hence $\lambda_1 \geq 1/4 - (7/64)^2 \approx 0.238$ for congruence covers of $\text{PSL}_2(\mathbb{Z}) \backslash \mathbb{H}^2$. Base-changing from $\text{GL}_2/\mathbb{Q}$ to GL_2/K would conditionally improve the Bianchi-side bound, but unconditional applicability to every X_N in our list is not asserted.

Remark 3.4 (Selberg strong is open). The Selberg eigenvalue conjecture (in its strong form, adapted to $\mathbb{H}^3$) predicts $\lambda_1 \geq 1$ for arithmetic congruence quotients. This is open. Numerically, the empirical match of Section 5 corresponds to the saturation case $\lambda_1 = 1$, taken as a working assumption motivated by the data.

3.3 The motivating prefactor $C(N)$

We define

$$C(N) := \sqrt{2\pi e} \sqrt{2/3} F(N), \quad F(N) = \frac{9}{10} \cdot \frac{N^2 + 1}{N^2}. \quad (2)$$

Numerically, $\sqrt{2\pi e} \simeq 4.1327$, $\sqrt{2/3} \simeq 0.8165$, $F(\infty) = 9/10$, $F(3) = 1$, giving $C(\infty) \simeq 3.037$ and $C(3) \simeq 3.374$.

The three factors are motivated as follows; *we do not claim a derivation from 4D Yang–Mills*.

- $\sqrt{2\pi e}$ is a Stirling-type constant arising in saddle-point analyses of Mellin transforms of heat traces on hyperbolic spaces. Its appearance here is motivated by the Minakshisundaram–Pleijel expansion [21, §4] and the Karamata tauberian theorem [13, 9], but the precise step linking the heat trace on X_N to a glueball mass on $\mathbb{R}^4$ is not rigorous.
- $\sqrt{2/3}$ is the value of $\xi^* = 1/(1 + |A_4/A_2|)$ where A_2, A_4 are Seeley–DeWitt coefficients on $\mathbb{H}^3$ [10, 21]. This is a universal property of constant negative curvature in dimension three, but its identification with the glueball-mass ratio in Yang–Mills is again not rigorous.
- $F(N) = (9/10)(N^2 + 1)/N^2$ is motivated by a Dijkgraaf–Witten genus expansion of the partition function of 2D Yang–Mills, with the normalisation fixed by setting $F(3) = 1$ for the constituent-gluon model. Whether this 2D-motivated formula has any rigorous origin in 4D $SU(N)$ Yang–Mills is an open question (Section 7).

Definition 3.5. The *arithmetic Hamiltonian* on $L^2(X_N)$ is

$$H_{\text{arith}}(N) := \sigma_0 C(N)^2 \Delta_{X_N},$$

where $\sigma_0 > 0$ is a fixed scale (we may take $\sigma_0 = 1$ in purely arithmetic normalisation; in the physical comparison $\sqrt{\sigma_0} = \sqrt{\sigma_{\text{lat}}}$, the lattice string tension).

Remark 3.6 (Cl[2]. vs. centre] We do not assert that $\text{Cl}(K)[2] \simeq Z(SU(N))[2]$ as groups; an earlier version of this work (now retracted internally) suggested such an isomorphism, which was caught as a fabrication: the correct correspondence runs through the Bianchi homology $H_1(X_N, \mathbb{Z}/2)$ (which carries the 2-torsion of $\text{Cl}(K)$ in its top degree) and the centre acts on Wilson loops via N -ality, not via a group isomorphism. The arithmetic content is captured by the dictionary K_{ASP} alone.

4 Empirical scaling relation

Proposition 4.1 (Tautological scaling). *Let $N \geq 2$ be such that $K_{\text{ASP}}(N)$ is defined, and let $X_N = \mathbb{H}^3/\text{PSL}_2(\mathcal{O}_{K_{\text{ASP}}(N)})$ and $H_{\text{arith}}(N) = \sigma_0 C(N)^2 \Delta_{X_N}$ on $L^2(X_N)$, with $C(N)$ defined in (2). Then, by construction,*

$$\frac{m_{\text{arith}}(N)}{\sqrt{\sigma_0}} = C(N) \sqrt{\lambda_1(X_N)}, \quad (3)$$

where $m_{\text{arith}}(N) := \sqrt{\sigma_0 C(N)^2 \lambda_1(X_N)}$. The identity (3) is tautological and asserts no lower bound on $\lambda_1(X_N)$.

Remark 4.2 (Status of any universal lower bound). Sarnak’s 1983 paper [16] establishes a Selberg-type lower bound $\lambda_1 \geq 21/25$ on certain specific arithmetic hyperbolic 3-manifolds. Whether the same bound applies to every Bianchi orbifold $\text{PSL}_2(\mathcal{O}_{K_{\text{ASP}}(N)}) \backslash \mathbb{H}^3$ in the anchor list of this work, for arbitrary class number $h_K = N$, is an open question; we do *not* invoke such a universal bound in any of the statements below. The Kim 2003 result [14] (Kim–Sarnak appendix) gives a Selberg approximation $\theta \leq 7/64$ on $\text{GL}_2/\mathbb{Q}$, which, by base change to GL_2/K , would conditionally improve λ_1 on the Bianchi side; unconditional applicability to every X_N in our list is not asserted. The empirical match of Section 5 corresponds numerically to the saturation case $\lambda_1(X_N) = 1$, a working assumption motivated by the data rather than by a proved bound.

5 Empirical evidence : the AT2021 lattice

5.1 Continuum-extrapolated lattice data

Athenodorou–Teper 2021 [1] (hep-lat/2106.00364) computed the continuum-extrapolated glueball spectrum of pure $SU(N)$ Yang–Mills theory on the lattice for $N \in \{2, 3, 4, 5, 6, \infty\}$. The

lowest scalar $m_{0^{++}}^{\text{lat}}/\sqrt{\sigma_{\text{lat}}}$ ratio (one-particle state, ground state of the 0^{++} channel) takes the values:

N	$m_{0^{++}}^{\text{lat}}/\sqrt{\sigma_{\text{lat}}}$ (AT2021)	$C(N)\sqrt{\lambda_1 = 1}$ (Prop. 4.1, $\lambda_1 = 1$ saturation)
2	3.781(23)	3.773
3	3.405(21)	3.374
4	3.271(27)	3.235
5	3.156(31)	3.170
6	3.102(32)	3.135
∞	3.07(?)	3.055

Table 2: Lattice data and $C(N)$ predictions assuming $\lambda_1(X_N) = 1$ (Selberg strong saturation). Root-mean-square deviation across 6 anchors: 0.85%.

5.2 Bayesian model comparison (under $\lambda_1 = 1$ saturation)

Caveat. The comparison below assumes saturation $\lambda_1(X_N) = 1$ for the structural model, which is a working assumption (not a proved bound). Under this assumption, the Bayesian comparison evaluates the structural shape of $C(N)$ against a free 2-parameter alternative; it does not test the saturation assumption itself, nor establish any physical identification of m_{arith} with the lattice glueball mass.

We compare two models on the 7 AT2021 anchors (we include the cross-channel 2^{++} and 1^{+-} anchors as auxiliary cross-checks, see [20, supplementary]):

Holy Grail (HG) (parameter-free): $m/\sqrt{\sigma} = C(N)$ with $C(N)$ given by (2), no free parameters.

Free (F) (two-parameter alternative): $m/\sqrt{\sigma} = \sqrt{2\pi e} \xi (\alpha(N^2+1)/N^2 + (1-\alpha))$ with ξ, α free.

The Bayesian Information Criterion (BIC) for $n_{\text{anchors}} = 7$ and Gaussian residuals of width $\sigma_{\text{lat}} \simeq 1\%$ gives:

$$\text{BIC}_{\text{HG}} = \chi_{\text{HG}}^2 + 0 \cdot \ln(7) = 5.62, \quad \text{BIC}_{\text{F}} = \chi_{\text{F}}^2 + 2 \cdot \ln(7) = 8.09.$$

$\Delta\text{BIC} = +2.47$, in the conventional reading equivalent to a Bayes factor $\text{BF}(\text{HG}/\text{F}) \simeq e^{2.47/2}$. prior ratio, which numerically evaluates to $\text{BF} \simeq 210$ for uniform priors on the box $\xi \in [0.5, 1]$, $\alpha \in [0, 1.5]$.

Remark 5.1 (What the Bayesian evidence supports). The Bayesian evidence supports the structural shape of $C(N)$ against any two-parameter alternative on the existing 7 anchors, with the proviso that no Bayesian comparison can establish a structural identification: it merely rules out two-parameter overfits.

5.3 Falsifiable predictions

1. **$N = 7$ test.** Proposition 4.1 under $\lambda_1 = 1$ gives $C(7) = \sqrt{2\pi e} \sqrt{2/3} (9/10)(50/49) \simeq 3.099$. Any future lattice continuum measurement outside $[3.09, 3.13]$ would refute the empirical formula.
2. **Cross-channel.** For $\xi_{2^{++}}^* = 2$ (heat-kernel spin-2 mode) and $\xi_{1^{+-}}^* = 3$, the corresponding $m_J/\sqrt{\sigma}$ inherits the multiplicative factors $\sqrt{2}$ and $\sqrt{3}$. Cross- N test against AT2021 currently gives RMS 2.24% (2^{++}) and 0.68% (1^{+-}).
3. **$K_{\text{ASP}}(32)$ existence.** A discovery that the admissibility set is empty at all scales for $N = 32$ would falsify the conjectural extension of the dictionary in its present form (Remark 2.4).

6 Open question: a transport principle

The empirical agreement of Section 5 is suggestive but, under Proposition 4.1 alone, $m_{\text{arith}}(N)$ is a quantity intrinsic to Bianchi geometry. We articulate the missing identification as an open question.

Remark 6.1 (Open transport question). Let $H_{\text{YM}}(N)$ denote the (conjectural) Hamiltonian of pure $\text{SU}(N)$ Yang–Mills theory on $\mathbb{R}^4$ in its lowest cuspidal sector (in the sense of the Yang–Mills Wightman functions, conditional on the existence of the Yang–Mills measure). Is there a unitary identification of low-lying spectra

$$\text{Spec } H_{\text{arith}}(N) \stackrel{\text{low}}{=} \text{Spec } H_{\text{YM}}(N),$$

in particular $m_{\text{arith}}(N) = m_{0++}^{\text{YM}}(\text{SU}(N))$, for every N in the anchor list? This question is open even before any precise formulation of the right-hand side. We do not claim such an identification.

We outline below four candidate routes that could in principle address the open transport question.

Route A : Arithmetic–gauge functor $\mathcal{F}$. Construct an explicit functor $\mathcal{F} : \text{Bianchi}_K \rightarrow \text{YM}_{\text{SU}(N)}$ sending the orbifold X_N to the moduli space of BPST instantons on $\mathbb{R}^4$ with c_2 controlled by class-field-theoretic data. The functor should intertwine Δ_{X_N} with the Lichnerowicz operator on instanton moduli (G1, Dirac index theorem twisted by genus characters). Estimated horizon: 5–10 years.

Route B : Selberg trace identity. Identify the Selberg trace $\text{Tr } e^{-t\Delta_{X_N}} = \sum_{\gamma} \ell(\gamma) e^{-\ell(\gamma)^2/4t} / \dots$ with the Feynman–Kac representation $\text{Tr } e^{-tH_{\text{YM}}} = \sum_{\text{inst}} e^{-S_E[\text{inst}]}$ in a controlled limit. The orbital integrals on X_N would match the instanton expansion. This is the route closest to Lemma A3-2 of this work [19, §5]. Estimated horizon: 6–12 years.

Route C : Langlands. Use Langlands functoriality to relate automorphic L -functions on GL_2/K (the Bianchi side) to gauge-theoretic correlators of $\text{SU}(N)$ via the geometric Langlands correspondence [8]. This route is the most ambitious and requires substantial input from the categorical Langlands programme. Estimated horizon: 10–20 years.

Route D : Bakry–Émery / CD inequality. Use the curvature-dimension inequality on X_N (which inherits a $\text{CD}(-1, \infty)$ structure from $\mathbb{H}^3$) to derive a Poincaré inequality with the same constant as Yang–Mills under a suitable parabolic regularisation. This routes via Holley–Stroock and Bakry–Émery [2]. Estimated horizon: 5–10 years.

All four routes are open. They form the explicit research programme attached to this work.

7 Open problems

We list the explicit open questions of the present work:

1. **(Q1) Uniform Selberg-type lower bound.** Does there exist a uniform positive lower bound $\lambda_1(X_N) \geq c > 0$ valid for every Bianchi orbifold X_N in the anchor list, with arbitrary class number? Sarnak’s 1983 paper [16] provides such a bound for specific arithmetic 3-manifolds; the applicability of its proof to every X_N in our list is open. Kim 2003 [14] would conditionally improve the bound on the cuspidal forms in the image of base change from $\text{GL}_2/\mathbb{Q}$.

2. **(Q2) $K_{\text{ASP}}(32)$.** Verify whether a fundamental discriminant $|D| \leq 10^9$ satisfies $h_K = 32$ and $\text{rk}_2 = 5$, or prove non-existence at all scales.
3. **(Q3) Derivation of $F(N)$.** Is there a rigorous derivation of the structural prefactor $F(N) = (9/10)(N^2 + 1)/N^2$ from 4D pure $\text{SU}(N)$ Yang–Mills? The current motivation is a 2D Dijkgraaf–Witten analogy.
4. **(Q4) Transport.** The open transport question of Remark 6.1: does the empirical match $m_{\text{arith}}(N) \approx m_{0++}^{\text{lat}}/\sqrt{\sigma_{\text{lat}}}$ admit a structural explanation through any of Routes A–D, or through another mechanism?
5. **(Q5) Constructive Yang–Mills measure on $\mathbb{R}^4$.** The Clay Millennium problem itself. Independent of the present work; required before the open transport question can be formally well-stated on the YM side.

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